Kali Linux on Raspberry Pi 5 TFT 3.5-inch Display Issue Resolution

This document provides a step-by-step guide to resolve the display issue with Kali Linux on a Raspberry Pi 5 using a 3.5-inch TFT display. The core of the problem was traced to an incorrect SD card directory reference within the display driver configuration, which required a change from `/dev/mmcblk0p2` to `/dev/sda2`.

Problem Description

Users attempting to run Kali Linux on a Raspberry Pi 5 with a 3.5-inch TFT display may encounter a situation where the display does not function correctly, often remaining blank or displaying artifacts. This issue typically arises due to the display driver looking for the system's root partition in the wrong location.

Solution Overview

The solution involves modifying the display driver configuration to correctly identify the root partition as `/dev/sda2` instead of `/dev/mmcblk0p2`. This ensures the display driver can properly initialize and interact with the display hardware. Specifically, the modifications will be made in the `cmdline.txt`, `cmdline.txt-noobs`, and `cmdline.txt-original` files located within the `/usr` directory of the `LCD-show-kali` folder.

Step-by-Step Guide

Follow these steps carefully to resolve the display issue:

Step 1: Access the Raspberry Pi 5

You will need to access your Raspberry Pi 5, either by connecting it to an external monitor and keyboard or by using SSH if you have network access.

Step 2: Navigate to the LCD-show-kali Folder

Open a terminal on your Raspberry Pi 5 and navigate to the `LCD-show-kali` folder. This folder is typically found in your home directory or where you extracted the display driver files.

For example, if it's in your home directory:
```
cd ~/LCD-show-kali
```

Step 3: Go to the /usr Directory

Once inside the `LCD-show-kali` folder, navigate to the `usr` subdirectory:
```
cd usr
```

Step 4: Backup the Original Configuration Files

Before making any changes, it is crucial to create backups of the original configuration files. This will allow you to revert to the previous state if any issues arise.
```
sudo cp cmdline.txt cmdline.txt.bak
sudo cp cmdline.txt-noobs cmdline.txt-noobs.bak
sudo cp cmdline.txt-original cmdline.txt-original.bak
```

Step 5: Edit the Configuration Files

Using a text editor like nano or vi, open each of the three configuration files (`cmdline.txt`, `cmdline.txt-noobs`, and `cmdline.txt-original`) one by one.

Edit cmdline.txt

Open the file:
```
sudo nano cmdline.txt
```

Within the file, search for any instance where the root partition is referenced as `root=/dev/mmcblk0p2`. You will need to change this reference to `root=/dev/sda2`.

For instance, you might find a line similar to:
`setenv bootargs "root=/dev/mmcblk0p2 rw rootwait console=tty1"`

You should change it to:
`setenv bootargs "root=/dev/sda2 rw rootwait console=tty1"`

Note: The actual parameter name and value might differ. The key is to identify the entry that specifies the root filesystem location and modify it to `/dev/sda2`.

Save the changes and exit the editor:

- **For Nano:** Press Ctrl+O to save, then Enter, and Ctrl+X to exit.

Edit cmdline.txt-noobs

Open the file:
`sudo nano cmdline.txt-noobs`

Perform the same `/dev/mmcblk0p2` to `/dev/sda2` modification as described for `cmdline.txt`.

Save the changes and exit the editor.

Edit cmdline.txt-original

Open the file:
`sudo nano cmdline.txt-original`

Perform the same `/dev/mmcblk0p2` to `/dev/sda2` modification as described for `cmdline.txt`.

Save the changes and exit the editor.

Step 6: Reboot the Raspberry Pi 5

To apply the changes, you need to reboot your Raspberry Pi 5.

```
sudo reboot
```

Step 7: Verify the Display

After the Raspberry Pi 5 reboots, the 3.5-inch TFT display should now be functioning correctly, displaying the Kali Linux interface.

Conclusion

By meticulously following these steps, the display issue experienced with Kali Linux on Raspberry Pi 5 using a 3.5-inch TFT display can be effectively resolved. The key was to correct the reference to the root partition within the display driver configuration files (`cmdline.txt`, `cmdline.txt-noobs`, and `cmdline.txt-original`) located in the `LCD-show-kali/usr` folder, ensuring the system correctly identifies and utilizes `/dev/sda2`.